

COURSE TITLE : PRINTED ELECTRONICS
COURSE CODE : 6106
COURSE CATEGORY : E
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Basic concept of printed electronics.	17
2	Scope of printed electronics.	20
3	Printing processes used for printed electronics	18
4	Future trends in printed electronics.	20
TOTAL		75

Course Outcomes:

SUB	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To know about printed electronics and its possibilities in the industry.
2	To understand about various printed electronics products.
3	To understand about different printing methods used in printed electronics.
4	To understand the opportunities and advancements in printed electronics.

SPECIFIC OUTCOMES

MODULE I BASIC CONCEPT OF PRINTED ELECTRONICS

1.1.0 To know about printed electronics and its possibilities in the industry.

- 1.1.1. To describe the concept of printed electronics.
- 1.1.2. To specify various printing methods used in printed electronics.
- 1.1.3. To explain the potential level of printed electronics in the industry.

MODULE II SCOPE OF PRINTED ELECTRONICS

2.1.0 To understand about various printed electronics products.

- 2.1.1 To specify the area of applications of printed electronics.
- 2.1.2 To explain the method of application of printed electronics.
- 2.1.3 To describe the benefits of printed electronics.

MODULE III PRINTING PROCESSES USED FOR PRINTED ELECTRONICS

3.1.0 To understand about different printing methods used in printed electronics.

- 3.1.1 To specify the chances of error during printed electronics production.
- 3.1.2 To explain the role gravure printing in printed electronics.
- 3.1.3 To explain the role of screen printing in printed electronics.
- 3.1.4 To describe digital printing methods used in printed electronics.

MODULE IV FUTURE TRENDS IN PRINTED ELECTRONICS.

4.1.0 To understand the opportunities and advancements in printed electronics.

- 4.1.1 To know various advanced technologies used in printed electronics production.
- 4.1.2 To specify different materials and their properties in printed electronics.
- 4.1.3 To define layer thicknesses of different printing methods.
- 4.1.4 To describe about the future market and future products of printed electronics.

CONTENT DETAILS

MODULE I

Introduction to printed electronics. Definition. Printed electronics production methods. Introduction to printing methods used in printed electronics. Need of printed electronics. Advantages of printed electronics. Comparison between printed electronics and conventional electronics. Potential of printed electronics. Application of printed electronics.

MODULE II

Advantages of printed electronics products from other electronic components – roll to roll production, mass production, flexible products, low price, etc. How printed electronic components are produced with printing process. Printing consideration for printed electronics - Resolution, registration, thickness, holes, materials. Benefits of roll to roll printing.

Printing process challenges – related printing of printed electronics. Material challenge – related printing of printed electronics. Manufacturing challenges. Device challenges.

MODULE III

Gravure printing for printed electronics. Uses of table – top proof presses and laboratory presses. Alignment process and source of error. Gravure printing considerations. Gravure printed products of printed electronics.

Screen printing for printed electronics. Flat screen and rotary screen. screen printing considerations. Screen printed products of printed electronics.

Flexographic printing for printed electronics.

Inkjet printing for printed electronics. Advantages and disadvantages of ink jet printing. Application of printed electronics.

MODULE IV

Aerosol printing process for printed electronics. Direct write extrusion process for printed electronics. Transfer printing process for printed electronics.

Printing materials for printed electronics. Different properties of materials. Organic and inorganic materials for printed electronics. Substrates for printed electronics. Conductor materials. Insulators materials. Semiconductor materials .

Printing technologies comparison based on layer thickness.

Future printed electronics market share. Future predicted products. OLED displays. Flexible displays.

Printed electronics sensors. Interactive packaging.

Printed electronics research companies and organizations.

Reference :		
Author	Title	Publishers
Katsuaki Suganuma	Introduction to Printed Electronics (Springer Briefs in Electrical and Computer Engineering)	Springer